THE VOLGA DELTA
The huge triangular delta of the Volga River is visible in the bottom of this image, with the Caspian Sea stretching out beyond it to the south.

ASIA *west*

Caspian Sea

LOCATION On the borders of Azerbaijan, Iran, Kazakhstan, Russia, and Turkmenistan

TYPE Saline inland sea

AREA 143,000 sq. miles (371,000 sq. km)

The Caspian Sea is the largest inland body of water on Earth. It contains salty rather than fresh water, so it can be appropriately described either as a salt lake or as an inland sea. The Caspian was once joined via another inland sea, the Black Sea, to the Mediterranean. However, several million years ago it was cut off from those other seas when water levels fell during an ice age. The sea has no outflow other than by evaporation, but it receives considerable inputs of water from the Volga River (supplying three-quarters of its inflow) and from the Ural, Terek, and several other rivers. Its surface level has changed throughout history in line with discharges from the Volga, which in turn have depended on rainfall levels in the Volga's vast catchment basin in Russia. Today, the Caspian Sea contains about 18,800 cubic miles (78,200 cubic km) of water—about one-third of Earth's inland surface water. Its salinity (saltiness) varies from 1 percent in the north, where the Volga flows in, to about 20 percent in Kara-Bogaz-Bol Bay, a partially cut-off area on its eastern shore.

OIL EXTRACTION
Some of Earth's largest oil reserves underlie the Caspian Sea. The greatest concentration of proven reserves and extraction facilities is in its northeastern section.

ANTARCTICA

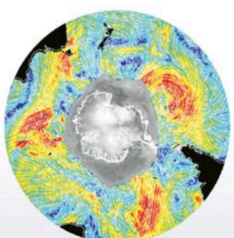
Antarctic Ice Sheet

LOCATION Covering most of Antarctica

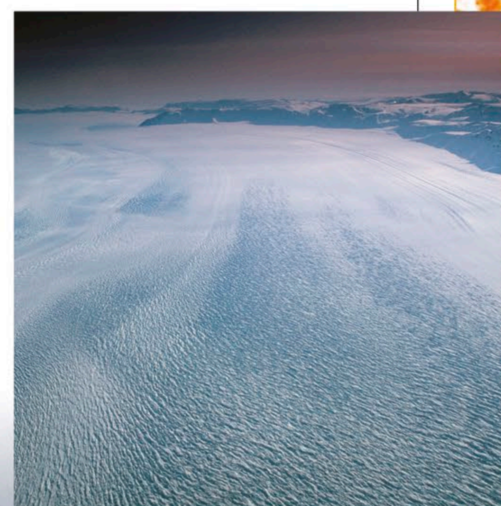
TYPE Continental ice sheet

AREA 5.3 million sq. miles (13.7 million sq. km)

Earth's largest glacier, the Antarctic Ice Sheet, is an immense mass of ice that covers almost all of the continent of Antarctica and holds over 70 percent of Earth's fresh water. The ice sheet has two distinct parts, separated by a range of mountains called the Transantarctic Range. The West Antarctic Ice Sheet has a maximum ice thickness of 2.2 miles (3.5 km), and its base lies mainly below sea level. The larger East Antarctic Ice Sheet is over 2.8 miles (4.5 km) thick in places with a base above sea level. Both parts of the ice sheet are domed, being slightly higher at their centers and sloping gently down toward their edges. A few areas around the edges of the ice sheets, such as some regions within the Transantarctic Range, are known to be rich sources of meteorites (see pp.220–221). Meteorites continually fall onto the ice sheet and become buried in it. But in a few places, where there is an upward flow of ice and some evaporation, they concentrate again at the surface. There are concerns that the West Antarctic Ice Sheet is shrinking due to global warming. Scientists agree that the West Antarctic Ice Sheet has been showing a general pattern of retreat for over 10,000 years, but think there is a small risk that it will collapse within the next few centuries as a result of climate change.



SATELLITE VIEW
This radar image shows the whole of Antarctica, with the larger, eastern section of its ice sheet on the left. The gray area around its coast is partly ice shelf and partly sea ice.



THE BEARDMORE GLACIER
This huge glacier drains the East Antarctic Ice Sheet into the Ross Ice Shelf. At 260 miles (415 km) in length, it is one of the longest glaciers on Earth.

THE LARSEN ICE SHELF
Around the coast of Antarctica, glaciers and ice streams merge to form platforms of floating ice called ice shelves. These are home to large colonies of penguins.